ANSWER KEY

Machine_Learning_Quiz2

Course: CSC354 - Machine Learning

CLO-2 Apply supervised learning techniques to solve classification problems.

Q1. Describe a scenario where applying supervised learning techniques to solve classification problems could significantly improve decision making in healthcare. Explain the potential benefits and challenges.

Answer:

In a hospital setting, supervised learning techniques can significantly improve decision making by accurately predicting patient readmission risks based on historical data and current health statuses. By training models with features such as previous admissions, medical history, lab results, and demographic information, the system could classify patients into high or low risk categories for readmission within 30 days of discharge. This would enable targeted interventions to reduce unnecessary hospital visits while optimizing resource allocation.

The potential benefits include improved patient outcomes through personalized care plans and reduced healthcare costs by avoiding preventable rehospitalizations. However, challenges may arise from data privacy concerns, the need for continuous model updates with new medical knowledge, ensuring diverse representation in training datasets to reduce bias, and maintaining interpretability of complex models.

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